

BABAKR HUSSAIN BABAKR SAAD

Software Developer & IT Professional

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[beko-cloud.work](#) · github.com/bekoblast · hub.docker.com/u/bekoblast

PROFESSIONAL SUMMARY

Software Developer and IT Professional with **17+ years** in IT systems, software engineering, and cloud technologies. Hands-on across full-stack web development, industrial IoT, ERP implementation, and modern frontend stacks (Next.js 16, React 19, TypeScript). Recent work includes rebuilding 2023-era industrial IoT dashboards from preserved Node-RED flows into modern React applications, publishing Dockerized SaaS products, and running GitHub-driven continuous deployment pipelines. Driven by clean architecture, dependable systems, and the discipline of automating away repeat work.

TECHNICAL SKILLS

Programming & Development

- JavaScript, TypeScript, Python, Node.js, PHP
- React 19, Next.js 16, Laravel, ASP.NET
- REST APIs, Express, Prisma

Modern Frontend & UI

- Tailwind CSS 4, shadcn/ui, Vite
- React Hook Form, TanStack Query, Zod
- Leaflet (maps), Recharts (data visualization), Lucide (icons)

AI & Modern Development Tools

- AI-assisted development (Claude Code, ChatGPT, Google Gemini, AI coding assistants)
- Prompt engineering and generative AI workflows
- AI tools for automation, development, and productivity

Systems & Cloud

- Linux server administration, Windows Server
- AWS Cloud Services, OpenStack
- Docker and containerization, Docker Hub image publishing
- Git version control, GitHub-driven workflows

DevOps & Continuous Deployment

- Netlify CI/CD pipelines wired to GitHub for auto-deploy on push
- Docker image build and registry publishing
- GitHub repositories, branches, and collaboration workflows

Databases

- MySQL, MariaDB, PostgreSQL, MongoDB, SQLite
- Database design, self-referencing hierarchical structures, migrations

IoT Development

- IoT systems architecture, sensor integration, data acquisition
- Binary protocol decoding (e.g. Tekelec ultrasonic sensor 140-byte packets, BCD encoding)
- Node-RED development, MQTT pipelines, rule chains
- ThingsBoard and OpenRemote platforms

- Industrial IoT monitoring solutions across fuel and energy verticals

ERP & Business Systems

- Odoo ERP implementation and functional configuration
- Daftra ERP administration (invoicing, equipment records, reports, users)
- Business process analysis, ERP customization, user training

Other Technical Skills

- WordPress development and maintenance
- Web hosting and cPanel
- System monitoring and troubleshooting
- Email systems administration

PROFESSIONAL EXPERIENCE

IT & Systems Coordinator

March 2024 - Present

Naif Obaid Al-Shammari Contracting Establishment - Saudi Arabia

- **Administer Daftra ERP system: invoicing, reports, and user management**
- Designed and shipped the JNCO Trading & Contracting corporate website (lovely-mochi-c24ebb.netlify.app)
- Designed and shipped the GCC Advanced Contracting corporate website (gcc-advanced.netlify.app) - clean static rebuild replacing a compromised WordPress install
- Manage internal digital workflows and business systems
- Support IT infrastructure and technical operations
- Implement digital solutions to improve operational efficiency

System Administrator & IoT Software Developer

Feb 2021 - Jun 2023 (ongoing freelance)

GTS Hi-Tech (formerly Gezira Telecom Solutions) - Sudan / Dubai / Juba

- **Developed software applications for IoT platforms and connected devices**
- Built systems for data collection and remote monitoring (Canar fuel tankers, SEDC transformers)
- Integrated IoT platforms with web applications and databases
- Performed testing and optimization for security and performance
- Implemented and supported Odoo ERP - installation, functional configuration, business process mapping
- Designed and shipped the GTS Hi-Tech corporate website (gts-hitech.com)
- Continuing freelance collaboration on web and IoT projects

IT Technical Support & System Administrator

Aug 2009 - Jan 2021

Gezira Hila Impex Co. Ltd - Sudan

- Provided enterprise IT support and infrastructure maintenance
- Managed servers, backups, and system updates
- Configured networks, user accounts, and security policies
- Implemented internal IT solutions and documentation systems
- Designed and shipped the Gezira Hila Impex corporate website (celadon-beijinho-928b7f.netlify.app)

KEY PROJECTS

Industrial IoT

Canar Fuel Tanker Monitor - Modern React Rebuild

Next.js 16 / React 19 / Tailwind 4 / Leaflet / Recharts · fuel-tanker-monitor.netlify.app · github.com/bekoblast/fuel-tanker-monitor

Faithful React rebuild of the 2023 industrial IoT system originally built on Node-RED + ThingsBoard for Canar (Sudan). Decodes 140-byte binary packets from Tekelec ultrasonic sensors - model code, BCD IMEI, alarm flags, battery, 28 historical readings - and visualizes the fleet on an OpenStreetMap layer with per-tank radial gauges, LCD bars, and charts. The ThingsBoard platform was lost during the 2023 conflict; the Node-RED flows survived and seeded this rebuild.

SEDC Transformer Monitor - Modern React Rebuild

Next.js 16 / React 19 / Tailwind 4 / Leaflet / Recharts · sedc-transformer-monitor.netlify.app · github.com/bekoblast/sedc-transformer-monitor

Companion IoT rebuild for the Sudanese Electrical Distribution Company. Monitors 3-phase distribution transformers across Khartoum with 24-field telemetry per device - voltages (L-N + L-L), currents, power factor, frequency, plus per-phase timeseries. Same modern architecture as the Canar rebuild.

Original Industrial IoT Deployments (Node-RED + ThingsBoard)

Canar Company and Sudanese Electricity Company · 2022 - 2023 · at GTS Hi-Tech

Fuel Tank Monitoring System - sensor-integrated IoT system measuring and monitoring fuel levels across storage tanks with real-time web dashboard. Electrical Transformer Monitoring - IoT-based system tracking current in power transformers with centralized monitoring through IoT platforms and web interfaces.

Enterprise & SaaS

ACC Next - Modern Accounting Platform

Next.js 16 / React 19 / Prisma / MariaDB / Tailwind / shadcn/ui · private deployment

Modern accounting platform replacing a legacy CodeIgniter system. Chart of accounts as a self-referencing tree, journal entries with line items, posting workflow, and financial reports (trial balance, account statement, daily note). Hierarchical account codes auto-generate as prefixes (01 -> 0101 -> 010101 -> 01010101).

Aqarati Property Manager

React / TypeScript / Vite / Express / SQLite · github.com/bekoblast/property-manager

Arabic RTL web application for managing company-owned real estate in Saudi Arabia - buildings, units, tenants, contracts, payments, and maintenance - with Excel and PDF export. React + TypeScript front-end backed by an Express + SQLite local API.

Goals Tracker - Dockerized Team Goals App

React / Express / Node.js / Docker · hub.docker.com/r/bekoblast/goals-tracker · github.com/bekoblast/goals-tracker

Team goals tracking application with Arabic RTL React UI, role-based access (admin / manager / employee), notifications, and backup/restore. Ships as a single 64 MB container exposing port 4174 with a health check endpoint. Frontend + Express API + JSON persistence bundled together.

Web Development

Babakr Saad Portfolio

HTML / CSS / JavaScript / Netlify · beko-cloud.work · github.com/bekoblast/portfolio

Personal portfolio with dual-mode profile switching (IT vs. Construction & Tendering), click-anchored radial-wipe theme transitions, animated particle network background, and a content layer that re-renders per mode. Continuous deployment from GitHub - every push goes live in seconds.

- **JNCO Trading & Contracting** - Saudi construction contractor corporate site (lovely-mochi-c24ebb.netlify.app)

- **GCC Advanced Contracting** - UAE-based contractor corporate site, clean static rebuild (gcc-advanced.netlify.app)
- **GTS Hi-Tech** - ICT solutions company across Dubai, Khartoum, and Juba (gts-hitech.com)
- **Gezira Hila Impex** - Sudan-based gum arabic exporter corporate site (celadon-beijinho-928b7f.netlify.app)
- **Nile Petroleum (Nilepet)** - Bilingual EN/AR corporate site for Sudan's national petroleum company (magenta-zuccutto-c5c8c9.netlify.app)

EDUCATION

Bachelor of Computer Science and Information Technology

Sudan University of Science and Technology - Sudan · 2010

CERTIFICATIONS

- **Google IT Support Professional Certificate** - 2025
- **Meta Back-End Developer Certificate** - 2025
- **Meta React Native Specialization** - 2025
- **Google AI Essentials** - 2025
- **Google Generative AI Learning Path** - 2025
- **Generative AI Leader (Google)** - 2025
- **AWS Cloud Technical Essentials** - 2025
- **Google IT Automation with Python** - 2025
- **EF SET English Certificate C2** - 2023
- **Outsource Inventory Management (Edraak)**
- **ICDL - International Computer Driving License** (4 modules)

LANGUAGES

- **Arabic** - Native
- **English** - Fluent (EF SET C2 certified)